

REED PUTNAM PRACTICE · FALL 2026

Invariants, Monovariants, Parity & Coloring

Week 2 · what a process can never change

Thu Oct 1 / Fri Oct 2

The toolbox (1)

Invariant

A quantity $I(\text{state})$ that **every allowed move leaves unchanged**. If $I(\text{start}) \neq I(\text{target})$, the target is unreachable. Candidates: a sum, a sum mod 2 or mod m , a product, a gcd, a permutation's sign, a color count.

- **Monovariant:** Φ **strictly** increases (or decreases) with every move, in a set with no infinite chain (e.g. $\mathbb{N}$). Then the process stops, and only where no move applies.
- **Reachability = invariant + construction:** the invariant rules states out; a construction (often **running the process backwards**) reaches the rest.

The toolbox (2)

- > **Coloring for tilings:** color the cells so every placement of every tile covers a prescribed number of each color; compare with the region's color counts.
- > **Parity:** two states differing in a parity-type invariant are never connected. A quantity changing by ± 1 each step differs from its start, after t steps, by a number with the parity of t .
- > **Extremal link:** a state minimizing a monovariant is one where no move applies: Week 1's extremal principle in dynamic form.

Example 1

Parity invariant

Seven cups stand upside down on a table. A move turns over exactly four cups. Can all seven end right side up?

→ work it on the board (2 min)

Example 2

Monovariant

Each vertex of a finite graph is colored red or blue. A move picks a vertex that has more neighbors of the opposite color than of its own color and recolors it. Show that the process terminates.

→ work it on the board (2 min)

Example 3

An invariant that appears after one move

Three piles have 5, 49, 51 stones. A move either merges two piles or splits a pile with an even number of stones into two equal piles. Can we reach 105 piles of one stone each?

→ work it on the board (3 min); let students guess the invariant

How to recognize it

- > Look for a **process**: moves, replacements, flipping, recoloring, a game in disguise.
- > Tells: “prove this stops”, “you cannot reach”, “independent of the choices”, “how many reachable”, “fewest tiles”.
- > Ask in order: what is **obviously preserved**? what is **monotone**? which **coloring** meets every tile the same way? Then **write it down** and verify move by move.

Pitfalls

- > A monovariant needs *strict* monotonicity *and* a well-founded set (not $\mathbb{R}$).
- > An invariant only proves *impossibility*; “for which states” still owes a construction.
- > Coloring: check *every* orientation and position of *every* tile.

Now you

Problems 1–4.

1–2 warm-ups · 3–4 easy Putnam · 5–6 take-home

35 minutes, groups of 2–3

Solutions posted after Friday.